

Fangyuan Lin

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EDUCATION

Columbia University

September 2025 - May 2030
New York, NY

- **Ph.D. in Statistics** Graduate School of Arts and Sciences
- **Research Interests:** Probability Theory and its applications, Information Theory and its applications
- **First Year Advisor:** Prof. Victor de la Peña

University of California, Berkeley

May 2020 – Aug. 2024
Berkeley, CA

Highest Honors

- **B.A. in Mathematics** College of Letters and Science
- **B.A. in Computer Science** College of Computing, Data Science, and Society
 - * **Dorothea Klumpke Roberts Prize in Mathematics:** “awarded to seniors who have demonstrated truly exceptional scholarship in mathematics, with a cash prize.” [\[link\]](#)
 - * **Outstanding Undergraduate Student Instructor Award** for exceptional teaching work. [\[link\]](#)

PUBLICATIONS & PREPRINTS

1. **A Selection Premium Decomposition for the Expected Maximum of Random Walks**
Victor H de la Pena, Fangyuan Lin, Victor K de la Pena (2026) ([preprint](#))
2. **An Analysis of Silk Density in Spider Webs**
Fangyuan Lin, Seewoo Lee, Jason Jiang, Grant Yang, Norman Sheu. (2025) ([Royal Society Open Science](#)).
3. **Revisiting the Unicity Distance through a Channel Transmission Perspective**
Fangyuan Lin (2024). *To appear in the PUMP Journal of Undergraduate Research.* ([arXiv](#))

TEACHING EXPERIENCE

STAT 5264: Stochastic Processes and Applications

Jan. 2026 – May. 2026
New York, NY

Teaching Assistant, Department of Statistics (Columbia University)

- [\[course page\]](#) on my personal website.

STAT 3104: Applied Bayesian Analysis

Sep. 2025 – Dec. 2025
New York, NY

Teaching Assistant, Department of Statistics (Columbia University)

- Support students with R and Stan implementations, maintain course materials and announcements, and assist with grading and feedback. [\[course page\]](#)

MATH 54: Linear Algebra & Differential Equations

Jan. 2024 – May 2024
Berkeley, CA

Teaching Assistant for Professor Zvezdelina Stankova (UC Berkeley)

- Taught 6 discussion sections per week and held 2 office hours, managing grading, proctoring, and administrative duties for a class of 56 students.
- Received exceptional feedback in official course evaluations and recognized as an **Outstanding Undergraduate Student Instructor**. [\[Teaching Evaluation\]](#)

MATH 1B: Calculus

Aug. 2023 – Dec. 2023
Berkeley, CA

Teaching Assistant for Dr. Norman Sheu (UC Berkeley)

- Taught 6 discussion sections per week and held 2 office hours, managing grading, proctoring, and worksheet creation for a class of 48 students.
- Received exceptional feedback in official course evaluations and recognized as an **Outstanding Undergraduate Student Instructor**. [\[Teaching Evaluation\]](#)

ACADEMIC AWARDS AND SCHOLARSHIPS

Dorothea Klumpke Roberts Prize in Mathematics [\[official site\]](#);

Department of Mathematics, UC Berkeley

Outstanding Undergraduate Student Instructor Award [\[official site\]](#);

The Graduate Division, UC Berkeley

Highest Honors in Mathematics, 2023-24;

Department of Mathematics, UC Berkeley

Summer Undergraduate Research Fellowship, 2023 [\[official site\]](#);

UC Berkeley

Phi Beta Kappa Honor Society

High Distinction in General Scholarship 2023-24;

College of Letters and Science, UC Berkeley

High Distinction in General Scholarship 2023-24;

College of Computing, Data Sci, & Society, UC Berkeley

TECHNICAL COURSEWORK

Doctorate Studies at Columbia University, 2025-2030

Statistics:

STAT 6101 *PhD Applied Statistics I* (A), STAT 6301 *PhD Probability Theory I* (A+)

Undergraduate Studies at UC Berkeley, 2020–2024:

Mathematics:

MATH 53 *Multivariable Calculus* (A), MATH 54 *Linear Algebra & ODE* (A+), MATH 55 *Discrete Math* (A),
MATH 74 *Intro to Upper-Div Math* (A+)
MATH 104 *Real Analysis* (A), MATH 106 *Measure-Theoretic Probability* (A+), MATH 110 *Abstract Linear Algebra*
(A+), MATH 113 *Abstract Algebra* (A), MATH 115 *Number Theory* (A-), MATH 124 *Programming in Math* (A+),
MATH 128A *Numerical Analysis* (A+), MATH 135 *Set Theory* (A), MATH 141 *Differential Topology* (A-), MATH
142 *Algebraic Topology* (A), MATH 160 *Math History* (A), MATH 185 *Complex Analysis* (A+), MATH 196 *Honors*
Thesis (A), MATH 199 *Independent Study*
MATH 202A *Graduate Analysis* (A-), MATH 205 *Graduate Complex Analysis* (A)

Statistics:

STAT 150 *Stochastic Processes* (A)

Computer Science:

CS 61A *Program Structures* (A), CS 61B *Data Structures* (A), CS 161 *Computer Security* (A), CS 171 *Cryptography*
(A), CS 188 *Artificial Intelligence* (A+)
EECS 126 *Probability & Random Processes* (A), EECS 127 *Convex Optimization* (A)

Physics:

PHYSICS 7A *Mechanics* (A+), PHYSICS 7B *Heat & Electricity* (A)

SKILLS & MISC.

- *Programming:* C, Golang, Java, Matlab, Julia, Mathematica, Python, R, Scheme, SQL, Swift
- *Machine Learning & Data Science:* TensorFlow, Pytorch, Sklearn, Scipy
- *Tools:* L^AT_EX, Anaconda, Git, Adobe Illustrator.
- *Hobbies:* Violin, piano, running, bodybuilding.

PH.D. PROGRAM MILESTONES

- Passed Ph.D. Qualification Exams upon arrival in September 2025.